

Garlock GS 8608



MATERIAL PROPERTIES*

Color:	Blue
Composition:	PTFE with inorganic fiber
Fluid Services¹:	Moderate concentrations of acids & alkalies, hydrocarbons, solvents, refrigerants and cryogenics
Temperature², °F (°C)	
Minimum:	-450°F (-268°C)
Continuous Max:	500°F (260°C)
Pressure, Maximum:	500 psig (34.5 bar)
P x T (max.)², psig x °F (bar x °C)	
1.5mm and 3.0mm:	150,000 psig x °F (5300 bar x °C)

TYPICAL PHYSICAL PROPERTIES*

ASTM F36	Compressibility, range, %:	10-20
ASTM F36	Recovery (min), %:	30
ASTM F38	Creep Relaxation, %:	58
ASTM D1708	Tensile (min), psi (N/mm²):	2000 (13.8)
	Elongation (min), %	240
ASTM D792	Specific Gravity:	2.1 - 2.3
ASTM F586	Design Factors	<u>1.5mm</u> <u>3.0mm</u>
	"m" factor:	3.5 3.5
	"y" factor, psi (N/mm ²):	4,000 (27.6) 7,000 (48.3)

SEALING CHARACTERISTICS*

	ASTM F37B Fuel A	ASTM F37B Nitrogen	DIN 3535- 4 Gas Permeability
Gasket Load, psi (N/mm²):	1000 (6.9)	3000 (20.7)	4640 (32)
Internal Pressure, psig (bar):	9.8 (0.7)	30 (2)	580 (40)
Leakage, Typical	0.5 ml/hr.	0.15 ml/hr.	0.015 cc/min

Notes:

This is a general guide and should not be the sole means of selecting or rejecting this material. ASTM test results in accordance with ASTM F-104; properties based on 1.5mm sheet thickness unless otherwise mentioned.

* Values do not constitute specification Limits

¹ See Garlock chemical resistance guide.

² Based on ANSI RF flanges at our preferred torque. When approaching maximum pressure, continuous operating temperature, minimum temperature or 50% of maximum P x T, consult Garlock Applications Engineering. Minimum temperature rating is conservative.

NOTE: Minimum recommended assembly stress = 3,600psi (25 N/mm²). Preferred assembly stress = 6,000-10,000psi (41-69 N/mm²). Gasket thickness of 1.5mm strongly preferred. Retorque the bolts/studs prior to pressurizing the assembly.